



# HORUS: Multimodal Large Language Models Framework for Video Retrieval at VBS 2025

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**Abstract.** In the dynamic field of video retrieval, precise and effective search methods are crucial for managing complex datasets. We present HORUS, a novel approach based on multimodal Large Language Models (mLLMs) that advances video retrieval capabilities through two key innovations: (1) advanced multi-modal feature aggregation, integrating text-to-image search with CLIP, free-text search from captions generated by Video-LLaMA2, and visual features from Video-LLaMA to capture temporal dynamics; and (2) GPT-based query expansion, combined with an advanced filter, addresses issues with low-quality open-ended text queries and refines item searches based on type and location within a scene. This work provides cutting-edge solutions for the VBS 2025 challenge and offers valuable insights into enhancing video search techniques.

**Keywords:** Video Browser Showdown · Video Retrieval · Multi-modal Feature Aggregation

## 1 Introduction

Video has become a dominant medium in the digital age, making effective content retrieval crucial. With the explosion of video content online, traditional search methods often fall short. Advanced retrieval technologies are essential for improving the accuracy and relevance of accessing video content.

At the International Conference on MultiMedia Modeling (MMM) 2025, the Video Browser Showdown (VBS) will feature tasks in Known-Item Search (KIS), Ad-Hoc Video Search (AVS), and Question Answering (Q/A), striving to locate specific video segments or answer questions based on visual or textual cues with

speed and accuracy. VBS 2025 will utilize a comprehensive dataset, including the expanded V3C collection [1], the Marine Video Kit (MVK) and its updated version, MVK 2.0 [2], and the LapGynLHE dataset for laparoscopic gynecology videos, to evaluate participants’ retrieval systems. The details of these datasets are shown in the Table 1.

**Table 1.** Details of the datasets in VBS 2025

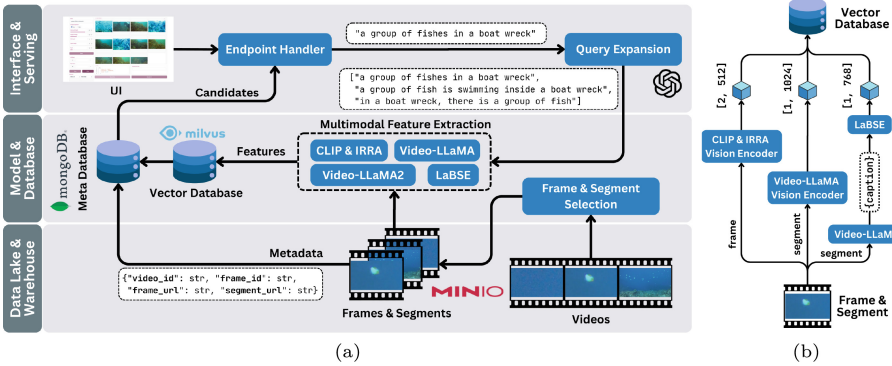
| Dataset Name      | # Videos | Size      | Task          |
|-------------------|----------|-----------|---------------|
| V3C [1]           | 28,450   | 8.7 TB    | KIS, AVS, Q/A |
| LapGynLHE         | 75       | 273.35 GB | KISV          |
| MVK & MVK 2.0 [2] | 3,193    | 501.01 GB | KISV          |

This work introduces HORUS, a framework that leverages multimodal Large Language Models (mLLMs) for enhanced video retrieval. Our contributions are twofold: (1) To address challenges in understanding and retrieving video segments, we aggregate feature representations from CLIP text-to-image search, free-text search from Video-LLaMA2 captions, and visual segment features from Video-LLaMA. This multi-modal aggregation improves our ability to capture temporal dynamics, significantly enhancing retrieval recall. (2) We tackle issues with low-quality, open-ended natural language queries, such as typos or misspellings, through GPT-based query expansion, improving query accuracy. Additionally, we integrate an advanced filtering mechanism that refines object searches based on object type and location within scenes, resulting in more precise and relevant search outcomes.

## 2 Related Work

At VBS 2024 [3], the 5th generation of VISIONE [4] emerged as the overall winner across multiple tasks, including KIS, AVS, and Q&A. Their success was attributed to enhancements in the user interface, which made it easier for novice users to perform searches. For similarity search, they retained the 4th version of VISIONE [5], which had been the runner-up at VBS 2023 [6]. This version allowed users to perform three distinct visual similarity searches, corresponding to different networks used for visual feature extraction DINOv2 [7], ALADIN [8], and CLIP2Video [9]. In the 5th version, VISIONE introduced an aggregated similarity search approach, leveraging late fusion to combine the ranked results from each method. They employed Reciprocal Rank Fusion [10] and further refined the top-5 results from each ranked list.

Vibro [11], not new to the VBS challenge, won in both 2022 [12] and 2023 [6]. Adhering to the philosophy of “never change a winning system,” they made no significant changes to their core approach between versions. Their system’s core technology for similarity search relied on the CLIP model for text-to-image retrieval. To rank the results, they used the FLAS [13] sorting algorithm, which improved ranking accuracy and sped up the sorting process.



**Fig. 1.** (a) Overview of HORUS comprising two flows: data ingestion and query processing. In the data ingestion, videos are processed to select keyframes and segments, followed by feature extraction. In the query processing, users submit queries through the UI, which are expanded and transformed into feature representations. Similarity scores are calculated, producing a ranked list of candidates. (b) Multimodal feature extraction for the visual track.

### 3 HORUS

HORUS is composed of three main sections: 1) Data lake and warehouse for managing videos, keyframes, and segments; 2) AI Models and Database conducting frame selection, feature extraction, and high-dimensional embedding searches; 3) User interface and serving for handling queries and applying filters to enhance search efficiency. Details are provided in Fig. 1(a).

#### 3.1 Data Lake and Warehouse

We propose an efficient method for constructing a retrieval system by leveraging MinIO<sup>1</sup> as a data lake to store and manage hundreds of gigabytes of video data.

#### 3.2 Model and Database

**Frame and Segment Selection.** Following the concept of “garbage in, garbage out”, we recognize that an effective database is crucial for the retrieval process, as objects that do not appear in the database cannot be found. To enhance retrieval efficiency, we focus on narrowing the search space. First, we apply the TransNetV2 model [14] to extract all events in the video. We merge adjacent segments incorrectly divided in the initial stage to prevent overly short clips, using the semantic similarity-based stitching technique from [15]. Lastly, we perform uniform sampling to select  $N$  frames per event as keyframes.

<sup>1</sup> <https://min.io>.

**Feature Extraction.** We present the multimodal feature extraction for the visual track in Fig. 1(b), which includes a set of CLIP-based models: CLIP and IRRa, which generate features from keyframes; a free-text search pipeline utilizing Video-LLaMA2 and LaBSE for extracting caption embeddings; and Video-LLaMA, which extracts visual representations from video segments.

**Text-to-Image.** The CLIP model [16] has seen extensive use across various research fields, owing to its ability to bridge visual concepts with natural language. To improve the model’s language recognition capabilities, we leverage the pre-trained Multi-lingual CLIP, as demonstrated by [17]. This model is robust and supports up to 100 languages, enabling it to easily understand various regional terms. For extracting visual representations, we use the CLIP visual encoder ViT-B/32, where  $v_{\text{CLIP}} = \Phi_{\text{CLIP-vis}}(I) \in \mathbb{R}^{1 \times 512}$ . Here, 512 denotes the embedding dimension,  $I$  is a given keyframe, and  $\Phi_{\text{CLIP-vis}}(\cdot)$  is the visual encoder of the CLIP model.

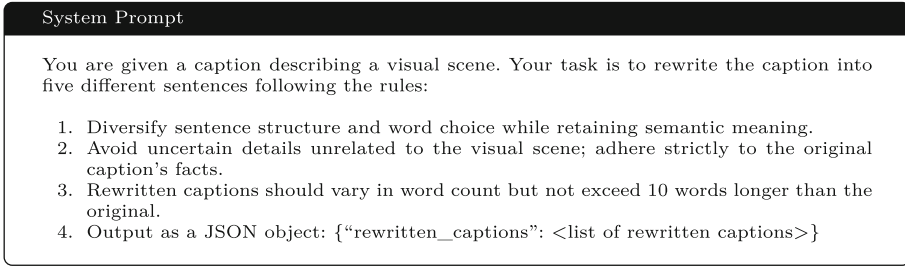
One significant limitation of the CLIP model is its tendency to misbind object attributes, even though it has been trained on extensive datasets [18]. This issue arises when attributes are incorrectly paired with objects in a sentence. To tackle this challenge, we use the pre-trained IRRa model from [19], which enhances CLIP with an implicit relation reasoning module to capture local tokens like object types and colors in both visual and textual inputs for accurate text-image matching. The visual encoder of IRRa model also use ViT-B/32, it generates a 512-dimensional embedding, where  $v_{\text{IRRA}} = \Phi_{\text{IRRA-vis}}(I) \in \mathbb{R}^{1 \times 512}$ .

**Free-Text Search.** We hypothesize that using the same modality for searching may yield better performance; therefore, we incorporate a free-text search module into our system. We begin by generating a caption for each segment using the pre-trained Video-LLaMA2 model [20]. Next, we use LaBSE [21] to encode the embeddings of these generated captions. LaBSE is a multilingual text encoder that generates effective sentence embeddings for 112 languages, enabling precise cross-lingual matching and retrieval of frames based on text descriptions in various languages. Given a caption for a certain segment  $C$ , the feature representation of the LaBSE model is  $c_{\text{LaBSE}} = \Phi_{\text{LaBSE}}(C) \in \mathbb{R}^{1 \times 768}$ . Here, 768 denotes the embedding dimension, and  $\Phi_{\text{LaBSE}}(\cdot)$  is the textual encoder of the LaBSE model.

**Text-to-Video.** While text-to-image retrieval has shown strong performance in image similarity search tasks, a significant challenge arises when transitioning from text-to-image to text-to-video retrieval due to the lack of temporal information between frames in most previous approaches. To bridge this gap, we utilize the pre-trained Video-LLaMA [22], a video understanding model equipped with a Q-former module. This module aggregates frame-level representations to effectively capture the temporal relationships between frames. The process starts by obtaining the segmented video, the Video-LLaMA model then generates a unified visual embedding for each segmented video chunk:  $v_{\text{Video-LLaMA}} = \Phi_{\text{Video-LLaMA-vis}}(S) \in \mathbb{R}^{1 \times 1024}$ . Here, 1024 denotes the embedding dimension,  $S$  is a given segmented video, and  $\Phi_{\text{Video-LLaMA-vis}}(\cdot)$  is the visual encoder of the Video-LLaMA model.

**Database and Vector Search.** We employ MongoDB<sup>2</sup> to store and query essential metadata and use this structured database to efficiently manage and investigate query requests.

For managing and searching high-dimensional vector data, we leverage Milvus<sup>3</sup>, an open-source vector database. To retrieve unified frames or segments using multimodal features, we apply the Weighted Scoring strategy. This approach assigns varying weights to results from each vector retrieval route based on the importance of each vector field. For instance, when searching for individuals dressed in specific clothing, the feature representation of IRRA may be deemed more crucial than other features. Let  $t_{\text{CLIP}}$ ,  $t_{\text{IRRA}}$ ,  $t_{\text{LaBSE}}$ , and  $t_{\text{Video-LLaMA}}$  denote the feature representations of the query input text provided by the end user. The similarity score function used here is cosine, so the matched scores between the query representations and vector embeddings in the database are  $s_{\text{CLIP}}$ ,  $s_{\text{IRRA}}$ ,  $s_{\text{LaBSE}}$ , and  $s_{\text{Video-LLaMA}}$ , respectively. The final weighted ranking SCORE is:  $s = w_{\text{CLIP}} \times s_{\text{CLIP}} + w_{\text{IRRA}} \times s_{\text{IRRA}} + w_{\text{LaBSE}} \times s_{\text{LaBSE}} + w_{\text{Video-LLaMA}} \times s_{\text{Video-LLaMA}}$ , where  $w_{\text{CLIP}} + w_{\text{IRRA}} + w_{\text{LaBSE}} + w_{\text{Video-LLaMA}} = 1$ .



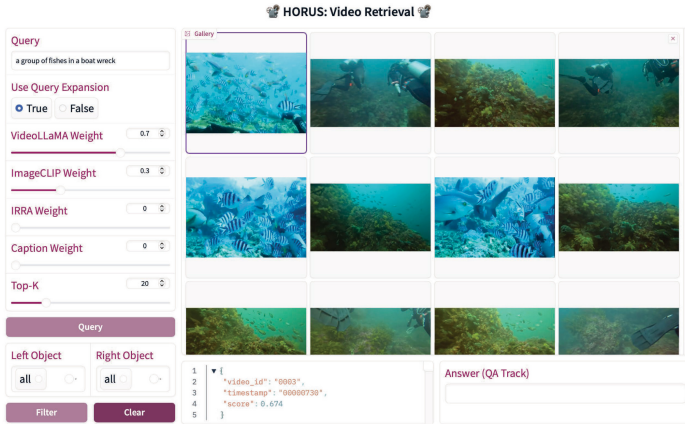
**Fig. 2.** Prompting the GPT for query expansion

### 3.3 Interface and Serving

**User Interface.** We use Gradio [23] to develop the user interface for HORUS as shown in Fig. 3. The interface is divided into three main areas: 1) Control Panel (Left Side): Includes a text box for entering queries and sliders for adjusting the weighted ranker. Below, there are two advanced filter boxes for selecting objects. 2) Search Results Gallery (Center): Displays the search results along with metadata for the matched frames. 3) Selected Frame Display (Below Gallery): Shows the metadata of the selected frame when clicked, along with an answer box for previewing possible answers in the Q/A track.

<sup>2</sup> <https://www.mongodb.com>.

<sup>3</sup> <https://milvus.io>.



**Fig. 3.** User Interface of HORUS

**Query Expansion and Advanced Filter.** At the outset of this work, we adhere to the principle of “garbage in, garbage out,” recognizing that poor query interpretation can lead to ambiguous search results. Query expansion is a common technique used to enhance the recall of search systems by adding additional terms to the original query [24]. Drawing on the techniques from [25] and [26] to tackle the challenge of ineffective open-ended text queries, we integrate a query expansion module that utilizes GPT<sup>4</sup> to rephrase the input query, the prompt employed is shown in Fig. 2.

We employ YOLOv8 [27] to identify objects in each keyframe. This acts as a filter, eliminating frames lacking the desired objects and locations, enhancing retrieval efficiency, and reducing processing time. For example, we aim to retrieve all frames containing a car located in the top-left area and ignore the rest.

## 4 Conclusion

We present HORUS, a framework that improves video retrieval through two key innovations. First, our advanced multi-modal feature aggregation utilizes CLIP for text-to-image search, free-text captions from Video-LLaMA2, and the Video-LLaMA model to capture dynamic temporal changes in visual scenes. Second, we integrate GPT-based query expansion with an advanced filtering mechanism, effectively tackling challenges posed by low-quality, open-ended text queries and refining searches by object type and scene location.

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<sup>4</sup> <https://openai.com/chatgpt/>.

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